MATHEMATICS · NUMBER

HCF and LCM: Highest Common Factor & Lowest Comm

Two essential tools — for simplifying, scheduling, and problem solving.

Use these notes well and you will be able to:

- DefineHCF and LCM and explain thedifference clearly.
- Find HCF using listing and prime factor methods.
- Find LCM using listing, prime factor, and formula methods.
- Choose the most efficient method for a given problem.
- Apply HCF and LCM to real-world word problems.

If you follow the steps, you will succeed.

Key Words

KEY WORD	SIMPLE DEFINITION
HCF	Highest Common Factor — the LARGEST number that divides exactly into two (or more) numbers.
LCM	Lowest Common Multiple — the SMALLEST number that is a multiple of two (or more) numbers.
Common factor	A factor shared by two or more numbers. e.g. 4 is a factor of both 12 and 20.
Common multiple	A multiple shared by two or more numbers. e.g. 12 is a multiple of both 4 and 6.
Prime factorisation	Writing a number as a product of primes. e.g. $36 = 2^2 \times 3^2$.
Intersection	The overlap — factors or primes that BOTH numbers share. Used for HCF.
Union	All factors or primes from BOTH numbers combined. Used for LCM.
Venn diagram method	Placing prime factors in overlapping circles to find HCF and LCM visually.
Lowest power	For HCF, take the LOWEST power of each common prime factor.
Highest power	For LCM, take the HIGHEST power of each prime factor from either number.

HCF — Three Methods

Method 1: Listing Factors — Find HCF of 24 and 36

NUMBER	FACTORS (list all, smallest to largest)
24	1, 2, 3, 4, 6, 8, 12, 24
36	1, 2, 3, 4, 6, 9, 12, 18, 36
COMMON	1, 2, 3, 4, 6, 12 HIGHEST is 12

Method 2: Prime Factors — Find HCF of 24 and 36

NUMBER	PRIME FACTORISATION	WORKING
24	$2^3 \times 3$	$24 = 2 \times 2 \times 2 \times 3$
36	$2^2 \times 3^2$	$36 = 2 \times 2 \times 3 \times 3$
HCF	$2^2 \times 3 = 12$	Take LOWEST power of each COMMON prime: 2^2 and 3^1

HCF(24, 36) = 12

Method 3: Venn Diagram — HCF of 24 and 36

24 only	BOTH	36 only
2^3 exclusive: 2	$2^2 \times 3$	3^2 exclusive: 3

HCF = product of OVERLAPPING (middle) section = $2^2 \times 3 = 12$

LCM — Three Methods

Method 1: Listing Multiples — Find LCM of 4 and 6

NUMBER	MULTIPLES (keep going until a match)
4	4, 8, 12, 16, 20, 24, 28, 32, …
6	6, 12, 18, 24, 30, …
LCM	First common multiple = 12

Method 2: Prime Factors — Find LCM of 24 and 36

NUMBER	PRIME FACTORISATION	WORKING
24	$2^3 \times 3$	$24 = 2 \times 2 \times 2 \times 3$
36	$2^2 \times 3^2$	$36 = 2 \times 2 \times 3 \times 3$
LCM	$2^3 \times 3^2 = 72$	Take HIGHEST power of EVERY prime: 2^3 and 3^2

LCM(24, 36) = 8 × 9 = 72

Method 3: Formula — LCM × HCF = Product of two numbers

RELATIONSHIP BETWEEN HCF AND LCM

$$\text{HCF} \times \text{LCM} = a \times b$$

where a and b are the two numbers.   Use when you know HCF and need LCM (or vice versa).

HCF vs LCM — When to Use Which

PROBLEM TYPE	USE	REASON
Splitting into equal groups — largest size	HCF	Largest divisor
Finding when two events coincide (repeat cycle)	LCM	Smallest shared multiple
Simplifying fractions (divide by...)	HCF	Divide by HCF
Adding/subtracting fractions (common denominator)	LCM	Use LCM as denominator
Sharing fairly — equal portions	HCF	HCF = portion size
Tilinga floor with equal tiles—largesttile	HCF	HCF of dimensions

Worked Examples — Real World

HCF problem: Two ribbons, 48 cm and 60 cm. Cut into equal pieces. Longest piece?

Find HCF(48, 60).
$48 = 2 \times 3$ | $60 = 2^2 \times 3 \times 5$
Common primes: 2^2 and 3. HCF = $4 \times 3 = 12$.
Longest equal piece = 12 cm. (48÷12=4 pieces, 60÷12=5 pieces)

LCM problem: Bus A runs every 12 min. Bus B every 18 min. When do they next meet?

Find LCM(12, 18).
$12 = 2^2 \times 3$ | $18 = 2 \times 3^2$
Take highest powers: 2^2 and 3^2. LCM = $4 \times 9 = 36$.
Both buses leave together again after 36 minutes.

Use the formula: HCF of two numbers is 8. Their product is 384. Find LCM.

Formula: HCF × LCM = $a \times b = 384$
LCM = 384 ÷ HCF = 384 ÷ 8
LCM = 48

Find HCF and LCM of 84 and 126

$84 = 2^2 \times 3 \times 7$
$126 = 2 \times 3^2 \times 7$
HCF: lowest powers of common primes (2, 3, 7): $2^1 \times 3^1 \times 7^1 = 42$
LCM: highest powers of ALL primes (2, 3, 7): $2^2 \times 3^2 \times 7 = 252$
HCF = 42 | LCM = 252   **Check: 42 × 252 = 10 584 = 84 × 126**

Practice Questions

HCF

1. Find HCF(18, 24) by listing factors.
2. Find HCF(120, 180) using prime factors.
3. The HCF of two numbers is 6. One number is 42. The other is between 30 and 50. Find it.

LCM

4. Find LCM(8, 12) by listing multiples.
5. Find LCM(30, 42) using prime factors.
6. Traffic lights A and B reset every 40 s and 60 s. When do they next reset together?

ADD YOUR NOTES HERE

Hey smart student!

HCF and LCM turnupeverywhere — fractions, algebra, real life.

The key: prime factorise both numbers first, then LOWEST for HCF, HIGHEST for LCM.

And always check: HCF × LCM = product of the two numbers.

Keep going — see you in the next lesson! From FREDi